

JAY CHOUKIKAR

Indore, India | +91 9302039813 | jaychoukiker@gmail.com | github.com/jayyyyqwq | linkedin.com/in/jay-choukikar

EDUCATION

IPS Academy, Institute of Engineering & Science (RGPV)

Indore, India

B.Tech in Computer Science and Engineering | CGPA: 8.2/10

2024 – 2028

Relevant coursework: Data Structures & Algorithms, DBMS, Machine Learning, Computer Networks, Operating Systems

Certifications: [AWS Certified AI Practitioner \(Sep 2026\)](#), [AWS Certified Cloud Practitioner \(Sep 2026\)](#)

EXPERIENCE

AI/ML Engineer Intern

May 2026 – Aug 2026

FireLLama Technology Pvt. Ltd.

Vadodara (Hybrid)

- Built a **document-parsing pipeline** using **MobileNet semantic segmentation** and **OCR** for structured field extraction.
- Fine-tuned on **5,000+ document images** with targeted augmentation, lifting **segmentation accuracy by 20+ points**.
- Optimized the inference pipeline for **3x faster** per-page processing on CPU.

PROJECTS

MaintainOps | *Serverless Predictive Maintenance on AWS*

SageMaker, Bedrock, Step Functions, Lambda, DynamoDB, AWS CDK, Python [GitHub](#)

- Built an end-to-end **AWS ML platform** that predicts industrial machine failures from live sensor data and generates **cited repair steps** from maintenance manuals via **Bedrock RAG** in ~10 s.
- Trained an **XGBoost** failure classifier on highly imbalanced data (3.4% failures), reaching **0.93 PR-AUC** and **85% precision / 86% recall**. Caught a validation bug that had silently cut precision from 83% to 49%.
- Automated retraining with **Step Functions** and a **quality gate** that blocks weaker models from production. Deploys in **one CDK command** at **<\$0.10/month idle**, backed by 128 CI tests.

ResHeightNet | *Paper Reimplementation: Single-Image Height Estimation*

PyTorch, ResNet34, GAMUS, pytest [GitHub](#)

- Reimplemented a published **ISPRS 2019 paper** (no official code) in **PyTorch**: a **ResNet34 encoder-decoder** that predicts building and tree heights from a **single aerial photo**.
- Achieved **2.17 m mean error** on 200 validation tiles, **34% lower** than a baseline assisted by the Depth Anything V2 foundation model, under the same evaluation protocol.
- Built a **pytest suite with 92%+ coverage**, including metric-parity checks and bitwise-verified checkpoint resume.

Truth Lens | *Android Food-Label Scanner*

Kotlin, Jetpack Compose, ML Kit, LLM Structured Outputs, Retrofit [GitHub](#)

- Built an **Android app** that scans a food barcode and returns a **personalized health score**, using **Open Food Facts (4M+ products)** or reading unknown labels with **on-device OCR + LLM**.
- Designed a **hybrid AI pipeline**: the LLM only extracts and explains, while scores and diet verdicts (Vegan / Jain / Fat Loss) come from **deterministic, tested engines**, so results never drift between runs.
- Recommended healthier swaps with **on-device embeddings**, with failure paths covered by **100 unit tests** in CI.

ContentGuard | *RL Environment for Copyright Decisions*

Python, FastAPI, OpenEnv, Docker, Hugging Face Spaces [GitHub](#)

- Built an **OpenEnv RL environment** where LLM agents issue **fair-use verdicts** under an **asymmetric reward**.
- Designed **14 archetypes across 7 tiers**, benchmarked **Llama 3.3 70B**, deployed on **HF Spaces (191 tests)**.

ACHIEVEMENTS

- 1st Prize**, IPS Intercollege Science Exhibition (Mar 2026): co-built **M-Ki Baat**, a wearable **sEMG brain-computer interface** that turns silent jaw clenches into speech using a **Random Forest** model.
- Top 100**, Meta x Scaler OpenEnv Hackathon (Apr 2026, 30,000+ registrations): built **AgentGrid**, a multi-agent LLM environment with **hardware-verified rewards**.

SKILLS

Languages: Python, Java, Kotlin, SQL, C++, Bash

ML / AI: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, Hugging Face Transformers, OpenCV, Pandas, NumPy

GenAI & CV: LLMs, RAG, Prompt Engineering, Semantic Segmentation, OCR

Cloud & MLOps: AWS (SageMaker, Bedrock, Lambda, Step Functions, DynamoDB, S3, CDK), Docker, CI/CD

Backend & Tools: FastAPI, Spring Boot, Apache Kafka, PostgreSQL, MySQL, GitHub Actions, Git, Linux